

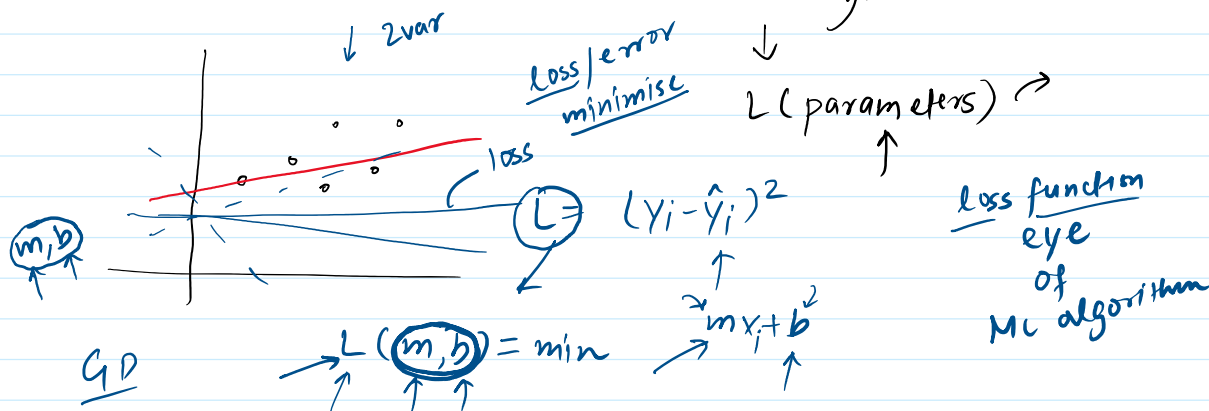
What is Loss Function?

23 March 2022 07:15

Loss function is a method of evaluating how well your algorithm is modelling your dataset.

high $\rightarrow$ poor
small $\rightarrow$ great

$$f(x) = x^2 + 2$$



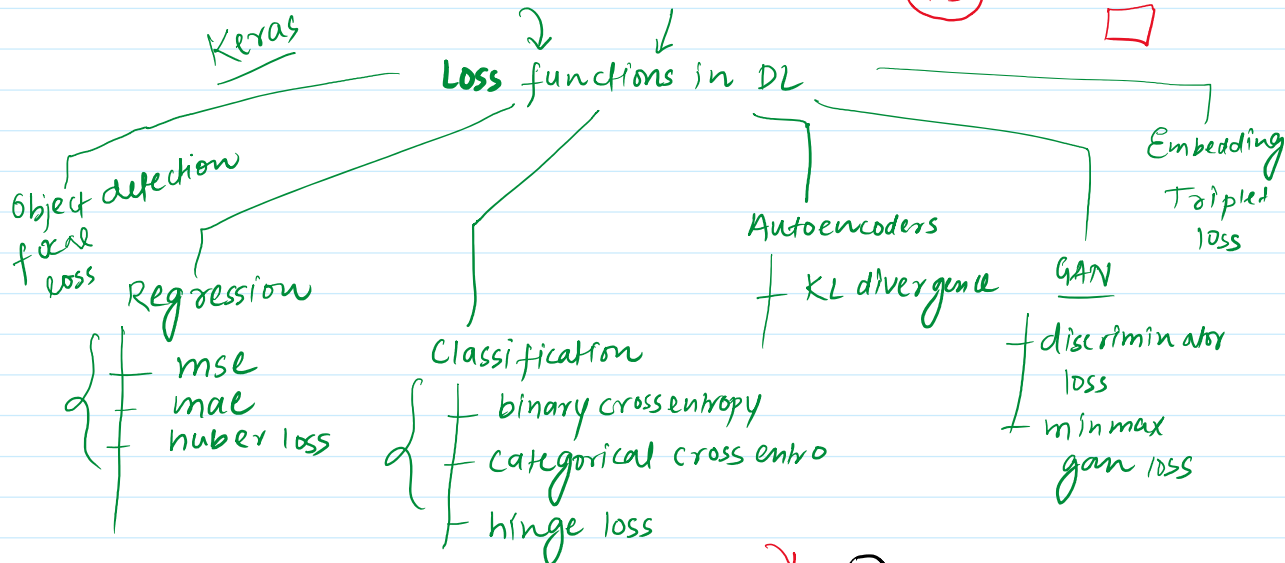
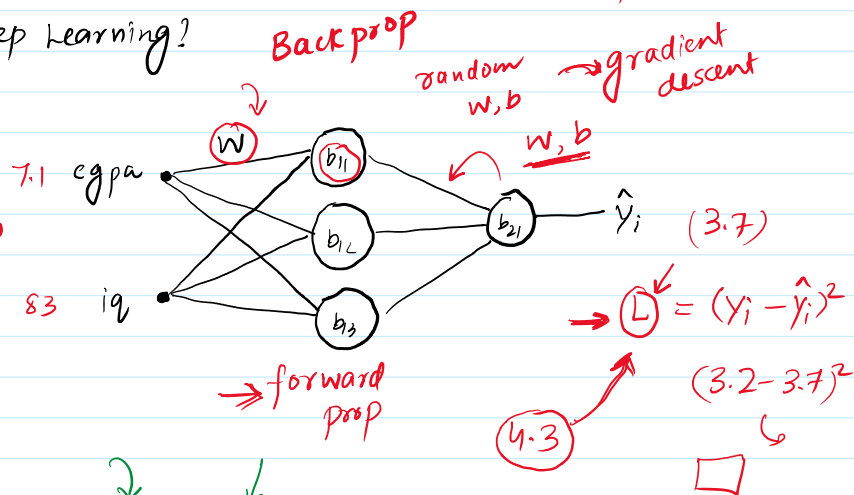
Why is Loss function important?

[You can't improve what you can't measure.]
Peter Drucker

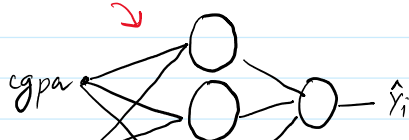
Loss Function in Deep Learning?

✓ cgpa | iq | package(eps)

7.1	83	3.2
8.5	91	4.5
6.3	102	6.1
5.1	87	2.7
...	...	...
...	...	...
...	...	...



Loss Function Vs Cost Function



-> By using the loss function what we do is:

- We choose weights and bias initially
- pass a row of data to the model
- it predict a number
- then we calculate the error using loss function
- then using this error we perform backpropagation
- we use gradient descent to update the w and b values
- then we move to another row and predict, calculate error, perform backprop, update w and b.
- we do this on every row with many epochs.
- finally we get perfect weights and biases that has the minimum loss
- and we get our perfect model that can predict perfect output based on external input.

-> Loss Function and Cost Function are not same

-> Loss function is the loss of each row

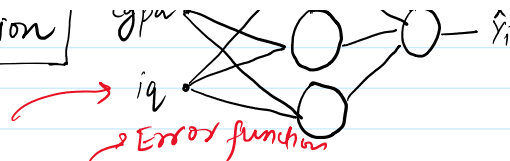
-> But cost function is universal. A number of cost for whole data.

-> e.g:

- Loss function: $(y_i - \hat{y}_i)^2$
- Cost function: $\frac{1}{n} \sum (y_i - \hat{y}_i)^2$

-> Loss function is also called Error Function.

Loss Function Vs Cost Function



Loss function → single training eg.

$$y_i = 6.3 \quad \hat{y}_i = 6.1$$

$$(y_i - \hat{y}_i)^2 = (6.3 - 6.1)^2 =$$

batch Cost function

$$\frac{1}{4} [(6.1 - 6.3)^2 + (4.1 - 4)^2 + (3.5 - 3.7)^2 + (7.2 - 7)^2] = CF$$

⑤ lprc $\sqrt{5}$

1. Mean Squared Error (MSE)

Squared loss L2 Loss

$$(y_i - \hat{y}_i)^2$$

(true - predict)²

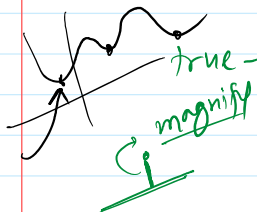
$$(6.3 - 6.1)^2 =$$

$$(y_i - \hat{y}_i)^2$$

$$(y_i - \hat{y}_i)^2$$

quadratic x^2

punish

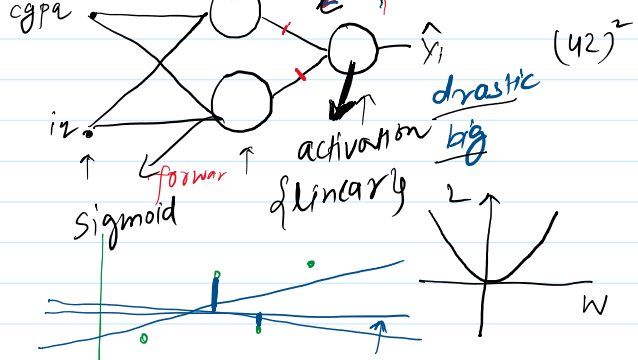


$$(y_i - \hat{y}_i)^2$$

$$(w, b) \rightarrow \text{min}$$

cgpa	iq	package	Prediction	error
6.3	100	6.3	6.1	0.2
7.1	91	4.1	4	0.1
8.5	83	3.5	3.7	-0.2
9.2	102	7.2	7	0.2
8.1	100	50	8	

Overall error



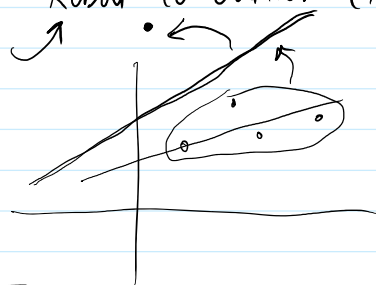
Advantages

- 1) Easy to interpret
- 2) Differentiable (GD)
- 3) 1 local minima

Disadvantage

- 1) Error unit (squared) → diff
- 2) Robust to Outliers (Not)

$$CF = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



2. Mean Absolute Error (MAE) → L1 loss

$$L = |y_i - \hat{y}_i|$$

$$C = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$CF = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- > We have already saw the advantages and disadvantages of mse and mae
- > If we have outliers on our data, we should use mae, if not then use mse.
- > Outlier is a big unusual data, that can interrupt the normal flow of model training.

$$C = \frac{1}{n} \sum_{i=1} |y_i - \hat{y}_i|$$

| true - predicted |
punish

Advantages

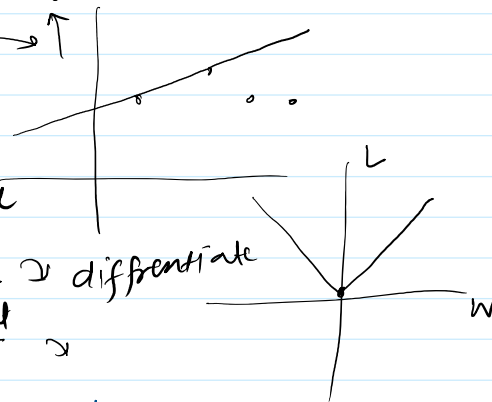
- 1) Intuitive and easy
- 2) Unit $\rightarrow$ same as y
- 3) Robust to outliers

Disadvantage

Not differentiable

GD $\rightarrow$ differentiate

Subgradient

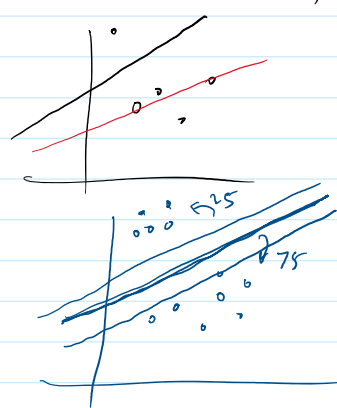


3. Huber Loss $\checkmark$

$$L = \begin{cases} \frac{1}{2} (y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta \\ \delta |y - \hat{y}| - \frac{1}{2} \delta^2 & \text{otherwise} \end{cases}$$

mae mse mse mae

mse - outliers $\checkmark$
mae - normal point $\checkmark$



4. Binary Cross Entropy $\rightarrow$ Log loss

- $\rightarrow$ classification
- $\rightarrow$ Two classes $\checkmark$

100 days

cgp a | iq | placement

8	80	1
7	70	0
6	60	0

$$\text{Loss function} = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$

Keras

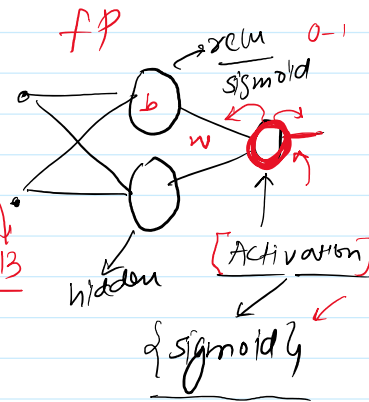
$$- (1-0) \log(1-0.25) - 1 \log(0.75)$$

$y \rightarrow$ actual value / target
 $\hat{y} \rightarrow$ NN prediction

$$\text{cost function} = -\frac{1}{n} \left[\sum_{i=1}^n y_i \log \hat{y}_i + (1-y_i) \log (1-\hat{y}_i) \right]$$

maximum likelihood

Logistic Reg



Advantage

Disadvantage

- * Differentiable $\rightarrow$ multi local minimum
- $\rightarrow$ Intuitive

5. Categorical Cross Entropy [used in Softmax Regression] $\rightarrow$ OHE

- > Huber Loss is the middle point of the mse and mae.
- > When there is no outliers it behaves like mse
- > Otherwise mae
- > We will get a data contains outliers or not by the hyper parameter delta.

-> We use binary cross entropy when the data is binary classification data.

-> When we use binary cross entropy, we must need to use sigmoid on the output layer. we can use relu/sigmoid on the hidden layer.

-> Output layer will predict between 0-1 by using sigmoid function

5. Categorical Cross Entropy [used in softmax regression]

→ [Multi-class] classification

1 point

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

where K is # classes in the data

1 point

$$L = - y_1 \log(\hat{y}_1) - y_2 \log(\hat{y}_2) - y_3 \log(\hat{y}_3)$$

$$[0.2 \ 0.3 \ 0.5]$$

$$[1 \ 0 \ 0]$$

$$L = -1 \times \log(0.2)$$

$$e^{z_1} + e^{z_2} + e^{z_3}$$

$$f(z) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

forward prop

softmax

$$[0.3 \ 0.6 \ 0.1]$$

$$[0 \ 1 \ 0]$$

$$-1 \times \log(0.6) = -$$

Sparse Categorical Cross Entropy

(L) ~

$$-1 \times \log(0.1)$$

{SCE, CCE} Loss function

SCE ~ fast
fast

$$-1 \times \log(0.4)$$

$$-1 \times \log(0.2)$$

$$[0.1 \ 0.4 \ 0.6]$$

$$[0.6 \ 0.2 \ 0.2]$$

(OHE) - fast
[SCE]

$$L = - \sum_{j=1}^K y_j \log(\hat{y}_j)$$

Cost function

$$C = - \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^K y_{ij} \log(\hat{y}_{ij})$$

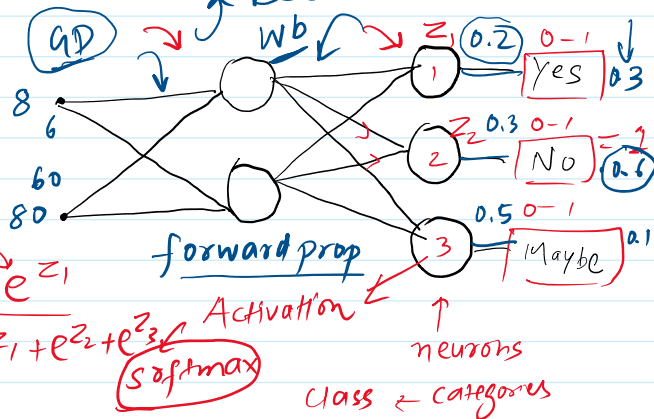
cgpa	iq
8	80
6	60
7	70

placed?

Yes 1
No 2
Maybe 3

Yes	No	Maybe
1	0	0
0	1	0
0	0	1

Yes
No
Maybe



- > If the data contains multiple class we use categorical cross entropy.
- > and it is used in softmax regression
- > on the output layer, we use softmax activation function.
- > Here the output layer contains that much node that the data has classes. if the data has 3 classes, the architecture must contain 3 node in output layer.
- > In sparse cat. cross entropy we don't do one hot encoding, here we do 1,2,3. Rest of the process are same.
- > If there is big number of classes we should use sce, cause it is fast. we do not need to do ohe on sce. also it don not calculate loss for extra classes rather than it's own class.

